Deep Generative Models

Dr. Amini

دانشگاه صنعتی شریف

department: Electrical Engineering

Kooshan Fattah Hesari 401102191

Problem Set 2

December 2, 2025

Deep Generative Models

Problem Set 2

Kooshan Fattah Hesari 401102191

Problem 1: Conditional Variational Autoencoder (CVAE)

Part 1: Derivation of the Objective Function

When training a Conditional Variational Autoencoder, our primary goal is to accurately capture the conditional distribution across the entire dataset. To achieve this, we establish a distance metric based on the expected Kullback-Leibler divergence, computed over the distribution of the conditioning variable y. More specifically, we aim to minimize the average divergence between the empirical data distribution $p_{\text{data}}(x|y)$ and our model's predicted distribution $p_\theta(x|y)$, where this average is weighted according to the frequency with which each conditioning value y appears in our dataset.

This leads us to define our loss function as:

$$\mathcal{L} = \mathbb{E}_{y\sim p_{\text{data}}(y)}\left[D_{\text{KL}}\left(p_{\text{data}}(x|y)\,||\,p_\theta(x|y)\right)\right]$$

To make this expression more concrete and computationally tractable, we expand the expectation operator and explicitly write out the definition of the Kullback-Leibler divergence. This gives us:

$$\mathcal{L} = \int p_{\text{data}}(y)\left(\int p_{\text{data}}(x|y)\log\left(\frac{p_{\text{data}}(x|y)}{p_\theta(x|y)}\right)dx\right)dy$$

We can simplify this nested integral by combining the integration operations:

$$\mathcal{L} = \iint p_{\text{data}}(y)\,p_{\text{data}}(x|y)\log\left(\frac{p_{\text{data}}(x|y)}{p_\theta(x|y)}\right)dx\,dy$$

Applying the fundamental chain rule of probability theory, we recognize that the joint distribution can be factored as the product of the marginal and conditional distributions: $p_{\text{data}}(x,y) = p_{\text{data}}(y)\,p_{\text{data}}(x|y)$. Substituting this relationship into our integral expression yields:

$$\mathcal{L} = \iint p_{\text{data}}(x,y)\log\left(\frac{p_{\text{data}}(x|y)}{p_\theta(x|y)}\right)dx\,dy$$

This double integral is precisely equivalent to computing the expectation with respect to the joint distribution over both x and y:

$$\mathcal{L} = \mathbb{E}_{(x,y)\sim p_{\text{data}}(x,y)}\left[\log\left(\frac{p_{\text{data}}(x|y)}{p_\theta(x|y)}\right)\right]$$

To derive our optimization objective, we expand the logarithmic ratio term using the properties of logarithms:

$$\mathcal{L} = \mathbb{E}_{(x,y)\sim p_{\text{data}}}\left[\log p_{\text{data}}(x|y)\right] - \mathbb{E}_{(x,y)\sim p_{\text{data}}}\left[\log p_\theta(x|y)\right]$$

The first term in this expression depends exclusively on the true underlying data distribution and remains constant regardless of our choice of model parameters θ. This term represents the conditional entropy of the data and cannot be reduced through optimization. Consequently, minimizing the overall distance $\mathcal{L}$ becomes equivalent to maximizing the second term, which represents the expected log-likelihood of our model.

Given a practical dataset consisting of N paired observations $\{(x_i, y_i)\}_{i=1}^N$ sampled from $p_{\text{data}}(x, y)$, we approximate the theoretical expectation using the Monte Carlo method with the empirical sample mean:

$$\text{Maximize} \quad J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N} \log p_\theta(x_i|y_i)$$

Part 2: Introduction of the Latent Variable

We now introduce the fundamental assumption that our data generation process involves a latent (unobserved) variable z. According to the rules of probability theory, particularly the sum rule (also known as marginalization), we can express the marginal likelihood $p_\theta(x|y)$ by integrating the joint distribution $p_\theta(x, z|y)$ over all possible configurations of the latent variable z.

Furthermore, by applying the chain rule of probability, we recognize that the joint distribution decomposes into the product of the likelihood function (decoder) $p_\theta(x|y, z)$ and the prior distribution $p_\theta(z|y)$. This decomposition gives us the integral representation:

$$p_\theta(x|y) = \int p_\theta(x, z|y)\, dz = \int p_\theta(x|y, z)\, p_\theta(z|y)\, dz$$

This equation provides a probabilistic interpretation: the probability of observing data point x given condition y equals the weighted average of the probability of observing x given both y and the latent variable z, where the weighting is determined by the probability of each latent configuration z given the condition y.

Part 3: Introducing the Variational Distribution

To make the inference problem tractable, we introduce an arbitrary conditional distribution $q(z|x, y)$, which will serve as our variational approximation to the true posterior. A critical requirement is that the support of $q(z|x, y)$ (the set of values where q is non-zero) must cover the support of the true posterior distribution. Under this assumption, we can multiply and divide the integrand by $q(z|x, y)$ without changing the value of the integral:

$$p_\theta(x|y) = \int p_\theta(x, z|y)\frac{q(z|x, y)}{q(z|x, y)}\, dz$$

Rearranging the terms to group the ratio of distributions, we obtain:

$$p_\theta(x|y) = \int q(z|x, y)\left(\frac{p_\theta(x, z|y)}{q(z|x, y)}\right)\, dz$$

By the fundamental definition of expectation, an integral of a function multiplied by a probability density function represents the expected value of that function under that distribution. Therefore, we can write:

$$p_\theta(x|y) = \mathbb{E}_{z\sim q(z|x,y)}\left[\frac{p_\theta(x, z|y)}{q(z|x, y)}\right]$$

Consequently, the expression for the quantity A that appears inside the mathematical expectation can be written as:

$$A = \frac{p_\theta(x, z|y)}{q(z|x, y)} = \frac{p_\theta(z|x, y)}{q(z|x, y)}\, p_\theta(x|y)$$

Part 4: Derivation of the Evidence Lower Bound (ELBO)

We begin our derivation with the log-marginal likelihood, also known as the model evidence, and substitute the expectation expression that we derived in the previous step:

$$\log p_\theta(x|y) = \log \left(\mathbb{E}_{z\sim q(z|x,y)} \left[\frac{p_\theta(x, z|y)}{q(z|x, y)} \right] \right)$$

At this point, we invoke Jensen's Inequality, a fundamental result in probability theory. Jensen's Inequality states that for any concave function ψ (such as the logarithm) and any random variable X, we have $\psi(\mathbb{E}[X]) \geq \mathbb{E}[\psi(X)]$. Applying this inequality to our expression yields:

$$\log p_\theta(x|y) \geq \mathbb{E}_{z\sim q(z|x,y)} \left[\log \left(\frac{p_\theta(x, z|y)}{q(z|x, y)} \right) \right] = \mathbb{E}_{z\sim q(z|x,y)} \left[\log \left(\frac{p_\theta(x|z, y)\, p_\theta(z|y)}{q(z|x, y)} \right) \right]$$

This lower bound on the log-likelihood is known as the Evidence Lower Bound (ELBO). We can further decompose the term inside the expectation to separate the reconstruction objective from the regularization term by using the properties of logarithms:

$$\text{ELBO}(x, y; \theta, q) = \mathbb{E}_{z\sim q(z|x,y)}[\log p_\theta(x|y, z) + \log p_\theta(z|y) - \log q(z|x, y)]$$

Breaking this into separate expectations:

$$\text{ELBO}(x, y; \theta, q) = \mathbb{E}_{z\sim q(z|x,y)}[\log p_\theta(x|y, z)] + \mathbb{E}_{z\sim q(z|x,y)} \left[\log \frac{p_\theta(z|y)}{q(z|x, y)} \right]$$

The second term can be recognized as the negative KL divergence between the variational posterior and the prior:

$$\text{ELBO}(x, y; \theta, q) = \mathbb{E}_{z\sim q(z|x,y)}[\log p_\theta(x|y, z)] - D_{\text{KL}}(q(z|x, y) \,||\, p_\theta(z|y))$$

Here, the first component represents the expected reconstruction quality (how well we can reconstruct x from z), while the second component (the KL divergence) serves as a regularizer that encourages the approximate posterior q to remain close to the prior p. This regularization is particularly important when the model will be used for generating new samples.

Part 5: Conditions for Tightness of the ELBO

Jensen's inequality transitions from an inequality to an equality if and only if the random variable inside the expectation is constant with respect to the distribution over which we are averaging. In our specific case, this requires that the ratio $\frac{p_\theta(x,z|y)}{q(z|x,y)}$ equals some constant value c for all values of z in the support of $q(z|x, y)$.

This constant ratio condition implies that $q(z|x, y)$ must be proportional to the joint distribution $p_\theta(x, z|y)$. The unique distribution that satisfies this proportionality constraint and integrates to 1 is the true posterior distribution:

$$q^\star(z|x, y) = p_\theta(z|x, y) = \frac{p_\theta(x|y, z)\, p_\theta(z|y)}{p_\theta(x|y)}$$

When $q(z|x, y)$ equals the true posterior distribution $p_\theta(z|x, y)$, the bound becomes tight, meaning the ELBO exactly equals the log-likelihood. The degree of tightness, or equivalently the gap between the true log-likelihood and the ELBO, can be quantified by examining their difference. We can derive this difference analytically through the following sequence of algebraic manipulations:

$$\log p_\theta(x|y) - \text{ELBO} = \log p_\theta(x|y) - \mathbb{E}_q\left[\log \frac{p_\theta(x, z|y)}{q(z|x, y)}\right]$$

Expanding the expectation:

$$= \mathbb{E}_q[\log p_\theta(x|y)] - \mathbb{E}_q\left[\log \frac{p_\theta(z|x, y)\, p_\theta(x|y)}{q(z|x, y)}\right]$$

Since $\log p_\theta(x|y)$ does not depend on z, it can be pulled out of the expectation:

$$= \mathbb{E}_q\left[\log p_\theta(x|y) - \log p_\theta(x|y) - \log \frac{p_\theta(z|x, y)}{q(z|x, y)}\right]$$

Simplifying:

$$= \mathbb{E}_q\left[\log \frac{q(z|x, y)}{p_\theta(z|x, y)}\right]$$

This is precisely the definition of the KL divergence:

$$\boxed{\log p_\theta(x|y) - \text{ELBO} = D_{\text{KL}}(q(z|x, y)\,||\,p_\theta(z|x, y))}$$

Thus, the gap between the true log-likelihood and the ELBO is exactly equal to the KL divergence between our approximate posterior q and the true posterior. This provides a precise measure of how well our variational distribution approximates the true posterior.

Part 6: The Optimization Problem and Its Challenges

The fundamental optimization problem in variational inference for a specific data point (x, y) is to find the variational distribution q that minimizes the KL divergence to the true posterior:

$$\min_q D_{\text{KL}}(q(z|x, y)\,||\,p_\theta(z|x, y))$$

However, solving this optimization problem directly presents two major computational challenges:

Challenge 1: Intractability of the True Posterior. The fundamental difficulty lies in the structure of the target distribution itself. The true posterior $p_\theta(z|x, y)$ contains the marginal likelihood $p_\theta(x|y)$ in its denominator, as shown by Bayes' rule:

$$p_\theta(z|x, y) = \frac{p_\theta(x|y, z)\, p_\theta(z|y)}{\int p_\theta(x|y, z)\, p_\theta(z|y)\, dz}$$

Computing the normalizing integral in the denominator (the evidence) is analytically intractable and computationally prohibitive for complex, high-dimensional models such as deep neural networks. The computational complexity grows exponentially with the dimensionality of the latent space, making direct evaluation impossible. This intractability prevents us from computing the KL divergence directly, as we cannot evaluate the target distribution.

Challenge 2: Scalability and Parameterization. Even if we could compute the true posterior, a naive approach to variational inference would be to assign a distinct set of variational parameters λ_i for each data point (x_i, y_i) in our dataset. We would then define $q(z|x_i, y_i; \lambda_i)$ and optimize λ_i individually for each sample.

This approach suffers from severe scalability limitations. First, the total number of parameters that need to be optimized grows linearly with the dataset size N. For a dataset containing millions of examples, we would need to maintain and optimize millions of separate parameter sets $\{\lambda_i\}_{i=1}^N$, which becomes computationally infeasible. Second, inference for a new, previously unseen data point would require running a complete iterative optimization procedure from scratch to find its optimal variational parameters λ. This would be far too slow for practical applications requiring real-time inference.

Solution: Amortized Inference with Neural Networks. To overcome both challenges simultaneously, Variational Autoencoders employ the technique of amortized inference. Rather than optimizing separate variational parameters for every individual data point, we learn a single global function—specifically, a neural network parameterized by ϕ—called the inference network or encoder. This network maps any input pair (x, y) directly to the parameters of the variational distribution:

$$z \sim q_\phi(z|x, y)$$

The encoder network takes x and y as inputs and produces the distributional parameters (such as the mean vector μ and variance σ^2) that characterize q. We then jointly optimize the ELBO with respect to both θ (the generative model parameters) and ϕ (the variational parameters) using stochastic gradient descent. This "amortizes" the cost of inference: we pay the computational cost once during training to learn the encoder function, and then inference for any new data point—whether from the training set or completely new—requires only a single forward pass through the encoder network. This makes the approach both scalable to large datasets and efficient for real-time applications.

Part 7: Computing the ELBO for New Data

When we encounter a new data pair $(x_{\text{new}}, y_{\text{new}})$ after training is complete, we need to compute the ELBO to evaluate the model's performance or to continue learning. The computation procedure consists of the following steps:

Step (a): Encoding. We feed the new data pair $(x_{\text{new}}, y_{\text{new}})$ into the trained encoder network q_ϕ to obtain the parameters of the variational distribution. For example, the encoder might output a mean vector μ_{new} and a variance vector σ^2_{new}, defining a Gaussian distribution $q_\phi(z|x_{\text{new}}, y_{\text{new}}) = \mathcal{N}(z; \mu_{\text{new}}, \sigma^2_{\text{new}} I)$.

Step (b): Sampling Latent Variables. To evaluate the expectation in the ELBO, we draw L independent samples $\{z^{(1)}, z^{(2)}, \ldots, z^{(L)}\}$ from the variational distribution. To enable gradient-based optimization through this sampling operation, we employ the reparameterization trick. Instead of sampling $z^{(l)}$ directly from $q_\phi(z|x_{\text{new}}, y_{\text{new}})$, we sample from a standard normal distribution $\epsilon^{(l)} \sim \mathcal{N}(0, I)$ and then transform it deterministically:

$$z^{(l)} = \mu_{\text{new}} + \sigma_{\text{new}} \odot \epsilon^{(l)}, \quad \text{where} \quad \epsilon^{(l)} \sim \mathcal{N}(0, I)$$

This reparameterization makes the sampling process differentiable with respect to the encoder parameters ϕ, allowing gradients to flow through the sampling operation during backpropagation.

Step (c): Estimating the ELBO. We compute an empirical Monte Carlo estimate of the ELBO using our L samples. The estimator is:

$$\text{ELBO} \approx \frac{1}{L}\sum_{l=1}^{L}\left(\log p_\theta(x_{\text{new}}|y_{\text{new}}, z^{(l)})\right) - D_{\text{KL}}(q_\phi(z|x_{\text{new}}, y_{\text{new}}) \,||\, p_\theta(z|y_{\text{new}}))$$

The first term averages the log-likelihood of reconstructing x_{new} across all sampled latent codes, while the second term (the KL divergence) can often be computed analytically when both the prior and variational posterior are Gaussian distributions.

It is important to note that if our goal is to strictly approximate the marginal likelihood $p_\theta(x_{\text{new}}|y_{\text{new}})$ rather than just its lower bound, the ELBO estimate may be overly conservative. A more accurate approximation of the marginal likelihood can be obtained using Importance Sampling with the variational posterior as the proposal distribution. The Importance Weighted estimator is given by:

$$p_\theta(x_{\text{new}}|y_{\text{new}}) \approx \frac{1}{K}\sum_{k=1}^{K}\frac{p_\theta(x_{\text{new}}|y_{\text{new}}, z^{(k)})\, p_\theta(z^{(k)}|y_{\text{new}})}{q_\phi(z^{(k)}|x_{\text{new}}, y_{\text{new}})}$$

where the samples $z^{(k)}$ are drawn from the variational posterior $q_\phi(z|x_{\text{new}}, y_{\text{new}})$. This importance-weighted estimator provides a tighter approximation to the true marginal likelihood, especially when using a larger number of samples K.

Part 8: Generating New Samples

To generate a new sample image corresponding to a given text description y_{new}, we utilize the generative components of the CVAE—specifically, the decoder network $p_\theta(x|y, z)$ and the prior distribution $p_\theta(z|y)$. Importantly, this generation process operates independently of the encoder network, since we are not conditioning on an observed x value. The complete generation procedure follows these steps:

Step (a): Sampling from the Prior. We begin by drawing a latent vector z from the prior distribution conditioned on the text description y_{new}. In many standard CVAE implementations, the prior is assumed to be a standard normal distribution that is independent of the conditioning variable, expressed mathematically as $z \sim \mathcal{N}(0, I)$. However, if our model architecture incorporates a learned conditional prior $p_\theta(z|y)$, we would instead sample from this learned distribution. The choice depends on the specific model design and training objectives.

Step (b): Combining Latent and Condition. We concatenate or otherwise combine the sampled latent vector z with the conditioning information y_{new}. The specific method of combination (concatenation, addition, or more complex fusion mechanisms) depends on the decoder architecture.

Step (c): Passing Through the Decoder. We feed this combined representation into the decoder network, which is parameterized as $p_\theta(x|y, z)$. The decoder processes this input through its neural network layers to produce the parameters of the output distribution.

Step (d): Obtaining Distribution Parameters. The decoder network outputs the parameters that define the distribution over the observable data x. For image generation tasks, these parameters might be:

- Pixel-wise probability distributions for discrete pixel values (e.g., categorical distributions over 256 intensity levels), commonly known as a Bernoulli distribution for binary images

- Parameters $(\mu_{\text{pixel}}, \sigma^2_{\text{pixel}})$ for a Gaussian distribution over continuous pixel values

Step (e): Generating the Final Output. To produce the final generated image x_{gen}, we have two options depending on the desired output characteristics:

- **Mean/Mode Output:** We can take the mean (for Gaussian) or mode (for categorical) of the output distribution. This typically produces smoother, more averaged results.
- **Sampled Output:** We can draw a random sample from the output distribution. This introduces additional stochasticity and can produce more varied results across multiple generations.

The complete generation process can be summarized as:

$z \sim p_\theta(z|y_{\text{new}}) \quad \text{or} \quad z \sim \mathcal{N}(0, I)$
Combine: $[z, y_{\text{new}}]$
Decode: $p_\theta(x|y_{\text{new}}, z)$
Output: $x_{\text{gen}} \sim p_\theta(x|y_{\text{new}}, z) \quad \text{or} \quad x_{\text{gen}} = \mathbb{E}_{p_\theta}[x|y_{\text{new}}, z]$

Problem 2: Cauchy–Schwarz Divergence

Part 1: Establishing the Fundamental Gaussian Product Property

Consider two Gaussian probability density functions: $p(x) = \mathcal{N}(x; \mu_1, \sigma_1^2)$ and $q(x) = \mathcal{N}(x; \mu_2, \sigma_2^2)$. The Cauchy–Schwarz divergence between these distributions is mathematically defined as:

$$D_{CS}(p||q) = -\log \frac{\int p(x)q(x)\,dx}{\sqrt{\int p(x)^2\,dx \int q(x)^2\,dx}}$$

To evaluate this divergence measure, we must first establish a fundamental property concerning the integral of the product of two Gaussian probability density functions. Our objective is to prove the following important result:

$$\int \mathcal{N}(x; \mu_1, \sigma_1^2)\mathcal{N}(x; \mu_2, \sigma_2^2)\,dx = \mathcal{N}(\mu_1; \mu_2, \sigma_1^2 + \sigma_2^2)$$

This result states that the integral of the product of two Gaussians evaluated over all possible values of x yields another Gaussian distribution evaluated at the means, with variance equal to the sum of the individual variances.

We begin by defining the integral explicitly in terms of the Gaussian density functions:

$$I = \int \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right) \cdot \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma_2^2}\right) dx$$

The first step in simplifying this integral is to combine the two exponential terms into a single unified exponential expression. We collect the constant normalization factors outside and focus on combining the exponents. Let us denote the combined exponent as E:

$$E = -\frac{1}{2}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(x-\mu_2)^2}{\sigma_2^2}\right]$$

To solve this integral analytically, we need to complete the square with respect to the variable x. This technique will allow us to recognize the integrand as another Gaussian distribution. We utilize a standard algebraic identity for combining two quadratic forms, which states that the sum of two normalized squared terms can be rewritten as:

$$\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(x-\mu_2)^2}{\sigma_2^2} = \frac{(x-\bar{\mu})^2}{\bar{\sigma}^2} + \frac{(\mu_1-\mu_2)^2}{\sigma_1^2+\sigma_2^2}$$

where the combined variance $\bar{\sigma}^2$ and the weighted mean $\bar{\mu}$ are defined through the following relationships:

$$\bar{\sigma}^2 = \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)^{-1} = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}$$

and the weighted mean (precision-weighted average) is:

$$\bar{\mu} = \bar{\sigma}^2\left(\frac{\mu_1}{\sigma_1^2} + \frac{\mu_2}{\sigma_2^2}\right) = \frac{\mu_1\sigma_2^2 + \mu_2\sigma_1^2}{\sigma_1^2+\sigma_2^2}$$

These expressions show that the combined variance is the harmonic mean of the individual variances (weighted by precision), and the combined mean is a precision-weighted average of the individual means.

Substituting this completed square form back into our expression for the exponent E:

$$E = -\frac{(x-\bar{\mu})^2}{2\bar{\sigma}^2} - \frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}$$

Now we can substitute this simplified exponent back into the original integral I:

$$I = \int \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(x-\bar{\mu})^2}{2\bar{\sigma}^2} - \frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right) dx$$

Using properties of exponentials, we can factor this as a product:

$$I = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right) \int \exp\left(-\frac{(x-\bar{\mu})^2}{2\bar{\sigma}^2}\right) dx$$

The integral term that remains is the kernel of an unnormalized Gaussian distribution with mean $\bar{\mu}$ and variance $\bar{\sigma}^2$. This integral evaluates to the normalization constant of a Gaussian:

$$\int \exp\left(-\frac{(x-\bar{\mu})^2}{2\bar{\sigma}^2}\right) dx = \sqrt{2\pi\bar{\sigma}^2} = \sqrt{2\pi} \cdot \frac{\sigma_1\sigma_2}{\sqrt{\sigma_1^2+\sigma_2^2}}$$

Substituting this result back into our expression for I:

$$I = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right) \cdot \sqrt{2\pi} \cdot \frac{\sigma_1\sigma_2}{\sqrt{\sigma_1^2+\sigma_2^2}}$$

Simplifying the constants:

$$I = \frac{1}{\sqrt{2\pi(\sigma_1^2+\sigma_2^2)}} \exp\left(-\frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right)$$

This is precisely the probability density function of a Gaussian distribution evaluated at μ_1 with mean μ_2 and variance $\sigma_1^2 + \sigma_2^2$:

$$\int \mathcal{N}(x; \mu_1, \sigma_1^2)\mathcal{N}(x; \mu_2, \sigma_2^2)\, dx = \mathcal{N}(\mu_1; \mu_2, \sigma_1^2 + \sigma_2^2)$$

With this fundamental result established, we can now proceed to compute all the necessary terms for the Cauchy-Schwarz divergence D_{CS}.

Computing the Numerator Term

The numerator of the Cauchy-Schwarz divergence is simply the result we just derived:

$$\int p(x)q(x)\, dx = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_2^2)}} \exp\left(-\frac{(\mu_1 - \mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)}\right)$$

Computing the Denominator Terms

For the denominator, we need to compute $\int p(x)^2\, dx$. We can use our general formula by setting $\mu_2 = \mu_1$ and $\sigma_2 = \sigma_1$. When both distributions are identical, the exponential term becomes $e^0 = 1$ since $(\mu_1 - \mu_1)^2 = 0$:

$$\int p(x)^2\, dx = \frac{1}{\sqrt{2\pi(2\sigma_1^2)}} = \frac{1}{2\sigma_1\sqrt{\pi}}$$

By the same reasoning, for the second distribution $q(x)$:

$$\int q(x)^2\, dx = \frac{1}{2\sigma_2\sqrt{\pi}}$$

The product of these two integrals inside the square root of the denominator is:

$$\int p(x)^2\, dx \int q(x)^2\, dx = \frac{1}{2\sigma_1\sqrt{\pi}} \cdot \frac{1}{2\sigma_2\sqrt{\pi}} = \frac{1}{4\pi\sigma_1\sigma_2}$$

Taking the square root of this product:

$$\sqrt{\int p(x)^2\, dx \int q(x)^2\, dx} = \frac{1}{2\sqrt{\pi\sigma_1\sigma_2}}$$

Forming the Ratio and Computing the Divergence

We now construct the ratio that appears inside the logarithm for D_{CS}:

$$\frac{\int pq}{\sqrt{\int p^2 \int q^2}} = \frac{\frac{1}{\sqrt{2\pi(\sigma_1^2+\sigma_2^2)}} \exp\left(-\frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right)}{\frac{1}{2\sqrt{\pi\sigma_1\sigma_2}}}$$

Simplifying this ratio:

$$= \frac{2\sqrt{\pi\sigma_1\sigma_2}}{\sqrt{2\pi(\sigma_1^2 + \sigma_2^2)}} \exp\left(-\frac{(\mu_1 - \mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)}\right)$$

Further simplification:

$$= \sqrt{\frac{2\sigma_1\sigma_2}{\sigma_1^2+\sigma_2^2}} \exp\left(-\frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right)$$

Finally, applying the negative logarithm to obtain the Cauchy-Schwarz divergence:

$$D_{CS}(p||q) = -\log\left(\sqrt{\frac{2\sigma_1\sigma_2}{\sigma_1^2+\sigma_2^2}}\right) - \left(-\frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right)$$

Using logarithm properties to expand:

$$D_{CS}(p||q) = -\frac{1}{2}\log\left(\frac{2\sigma_1\sigma_2}{\sigma_1^2+\sigma_2^2}\right) + \frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}$$

This can be rewritten in a more standard form:

$$D_{CS}(p||q) = \frac{1}{2}\log\left(\frac{\sigma_1^2+\sigma_2^2}{2\sigma_1\sigma_2}\right) + \frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}$$

Proving the Inequality with KL Divergence

Now we wish to demonstrate that $D_{CS}(p||q) \leq \min\{D_{KL}(p||q), D_{KL}(q||p)\}$.

First, let us observe a crucial symmetry property of the closed-form expression we derived for $D_{CS}(p||q)$:

$$D_{CS}(p||q) = \frac{1}{2}\log\left(\frac{\sigma_1^2+\sigma_2^2}{2\sigma_1\sigma_2}\right) + \frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}$$

Notice that this expression exhibits invariance under the exchange of parameter pairs (μ_1, σ_1) and (μ_2, σ_2). Specifically:

- The mean difference term $(\mu_1-\mu_2)^2$ is inherently symmetric
- The ratio $\frac{\sigma_1^2+\sigma_2^2}{2\sigma_1\sigma_2}$ remains unchanged when we swap σ_1 and σ_2

Therefore, we have the symmetry property:

$$D_{CS}(p||q) = D_{CS}(q||p)$$

To prove the main inequality $D_{CS}(p||q) \leq \min\{D_{KL}(p||q), D_{KL}(q||p)\}$, it suffices to show that $D_{CS}(p||q) \leq D_{KL}(p||q)$. Once this is established, the symmetry property immediately implies that $D_{CS}(q||p) \leq D_{KL}(q||p)$ also holds, which means D_{CS} is bounded above by both KL divergences.

The KL divergence between two Gaussian distributions is given by the standard formula:

$$D_{KL}(p||q) = \log\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2+(\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

Let us define $\Delta = D_{KL} - D_{CS}$ as the difference between the KL divergence and the Cauchy-Schwarz divergence. We can decompose this difference into two separate components—one involving the means and one involving the variances:

$$\Delta = \underbrace{\left(\frac{(\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{(\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right)}_{\Delta_\mu} + \underbrace{\left(\log\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2}{2\sigma_2^2} - \frac{1}{2} - \frac{1}{2}\log\frac{\sigma_1^2+\sigma_2^2}{2\sigma_1\sigma_2}\right)}_{\Delta_\sigma}$$

Analyzing the Mean Component Δ_μ:

We can rewrite Δ_μ by factoring out the common squared difference:

$$\Delta_\mu = \frac{(\mu_1 - \mu_2)^2}{2}\left(\frac{1}{\sigma_2^2} - \frac{1}{\sigma_1^2 + \sigma_2^2}\right)$$

To determine the sign of this expression, we need to compare the two fractions. Finding a common denominator:

$$\frac{1}{\sigma_2^2} - \frac{1}{\sigma_1^2 + \sigma_2^2} = \frac{(\sigma_1^2 + \sigma_2^2) - \sigma_2^2}{\sigma_2^2(\sigma_1^2 + \sigma_2^2)} = \frac{\sigma_1^2}{\sigma_2^2(\sigma_1^2 + \sigma_2^2)}$$

Since all variances are strictly positive quantities by definition (i.e., $\sigma_1^2 > 0$ and $\sigma_2^2 > 0$), we have:

$$\sigma_1^2 + \sigma_2^2 > \sigma_2^2 \implies \frac{1}{\sigma_2^2} > \frac{1}{\sigma_1^2 + \sigma_2^2}$$

Therefore, $\Delta_\mu \geq 0$ for any choice of means μ_1, μ_2 and variances σ_1, σ_2.

Analyzing the Variance Component Δ_σ:

The remaining terms are:

$$\Delta_\sigma = \left(\log\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2}{2\sigma_2^2} - \frac{1}{2}\right) - \left(\frac{1}{2}\log\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1\sigma_2}\right)$$

To simplify the analysis, let us introduce the substitution $t = \frac{\sigma_1}{\sigma_2}$, which represents the ratio of standard deviations. Then $\frac{\sigma_2}{\sigma_1} = \frac{1}{t}$. We can rewrite the entire expression in terms of this single variable t:

$$\Delta_\sigma(t) = -\log t + \frac{t^2}{2} - \frac{1}{2} - \frac{1}{2}\log\left(\frac{t^2\sigma_2^2 + \sigma_2^2}{2t\sigma_2^2}\right)$$

Simplifying the logarithmic term:

$$\Delta_\sigma(t) = -\log t + \frac{t^2}{2} - \frac{1}{2} - \frac{1}{2}\log\left(\frac{t^2 + 1}{2t}\right)$$

Expanding the logarithm using properties $\log(a/b) = \log a - \log b$ and $\log(ab) = \log a + \log b$:

$$\Delta_\sigma(t) = -\log t + \frac{t^2}{2} - \frac{1}{2} - \frac{1}{2}\log(t^2 + 1) + \frac{1}{2}\log 2 + \frac{1}{2}\log t$$

Combining like terms:

$$\Delta_\sigma(t) = \frac{t^2 - 1}{2} - \frac{1}{2}\log t - \frac{1}{2}\log(t^2 + 1) + \frac{1}{2}\log 2$$

To find the minimum of this function, we compute the derivative with respect to t:

$$\frac{d\Delta_\sigma}{dt} = t - \frac{1}{2t} - \frac{1}{2}\cdot\frac{2t}{t^2 + 1} = t - \frac{1}{2t} - \frac{t}{t^2 + 1}$$

Combining these terms over a common denominator $2t(t^2 + 1)$:

$$\frac{d\Delta_\sigma}{dt} = \frac{2t^2(t^2 + 1) - (t^2 + 1) - 2t^2}{2t(t^2 + 1)} = \frac{2t^4 + 2t^2 - t^2 - 1 - 2t^2}{2t(t^2 + 1)} = \frac{2t^4 - t^2 - 1}{2t(t^2 + 1)}$$

Setting the derivative equal to zero to find critical points: $2t^4 - t^2 - 1 = 0$.

Let $u = t^2$ (where $u > 0$ since t is a ratio of positive standard deviations):

$$2u^2 - u - 1 = 0$$

This factors as:

$$(2u + 1)(u - 1) = 0$$

The solutions are $u = 1$ and $u = -0.5$. Since $u = t^2$ must be positive (as $t > 0$), the only valid solution is $u = 1$, which implies $t = 1$.

Checking the value at $t = 1$:

$$\Delta_\sigma(1) = \frac{1-1}{2} - 0 - \frac{1}{2}\log(2) + \frac{1}{2}\log 2 = 0$$

To verify this is indeed a minimum, we examine the sign of the derivative. For $t < 1$, the numerator $2t^4 - t^2 - 1 < 0$, making the derivative negative. For $t > 1$, the numerator becomes positive. Therefore, the derivative changes sign from negative to positive at $t = 1$, confirming this is a global minimum.

Since $\Delta_\sigma \geq 0$ (with equality at $t = 1$, i.e., when $\sigma_1 = \sigma_2$) and $\Delta_\mu \geq 0$ for all parameter values, we conclude that the total difference is non-negative:

$$\Delta = \Delta_\mu + \Delta_\sigma \geq 0 \implies D_{KL}(p||q) - D_{CS}(p||q) \geq 0 \implies D_{CS}(p||q) \leq D_{KL}(p||q)$$

By symmetry, this also establishes that $D_{CS}(p||q) \leq D_{KL}(q||p)$, completing the proof of the desired inequality.

Part 2: Analysis of the Modified VAE Objective

We are given the following modified objective function for training a Variational Autoencoder:

$$\mathcal{L}_{CS}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \lambda D_{CS}(q_\phi(z|x)||p_\theta(z))$$

To understand how this objective relates to the standard VAE optimization framework, we must first recall the fundamental identity that governs Variational Autoencoders. The log-marginal likelihood (evidence) admits the following decomposition:

$$\log p_\theta(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x)||p_\theta(z)) + D_{KL}(q_\phi(z|x)||p_\theta(z|x))$$

where the last term represents the KL divergence between the approximate posterior and the true posterior.

From this fundamental decomposition, we can isolate the reconstruction term (expected log-likelihood) by rearranging:

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] = \log p_\theta(x) + D_{KL}(q_\phi(z|x)||p_\theta(z)) - D_{KL}(q_\phi(z|x)||p_\theta(z|x))$$

Now, substituting this expression for the reconstruction term into the definition of our modified objective $\mathcal{L}_{CS}$:

$$\mathcal{L}_{CS} = [\log p_\theta(x) + D_{KL}(q_\phi(z|x)||p_\theta(z)) - D_{KL}(q_\phi(z|x)||p_\theta(z|x))] - \lambda D_{CS}(q_\phi(z|x)||p_\theta(z))$$

Rearranging terms to group related components:

$$\mathcal{L}_{CS} = \log p_\theta(x) - \underbrace{D_{KL}(q_\phi(z|x)||p_\theta(z|x))}_{\text{Variational Gap}} + \underbrace{D_{KL}(q_\phi(z|x)||p_\theta(z)) - \lambda D_{CS}(q_\phi(z|x)||p_\theta(z))}_{\text{Regularization Difference}}$$

- $\log p_\theta(x)$**:** This is the actual model evidence (marginal log-likelihood) that we ultimately wish to maximize. It represents how well our model explains the observed data.

- $D_{KL}(q_\phi(z|x)||p_\theta(z|x))$**:** This term quantifies the approximation error—specifically, how well the variational posterior approximates the true posterior distribution. The learning process inherently minimizes this gap by optimizing the variational parameters, making the approximate posterior more accurate.

- $D_{KL}(q||p) - \lambda D_{CS}(q||p)$**:** This represents a modified regularization constraint that differs from the standard VAE formulation. In a standard VAE, we would simply subtract the KL divergence $D_{KL}(q||p)$, which encourages the approximate posterior to match the prior. Here, we are instead adding back a scaled version of the difference between the KL divergence and the Cauchy-Schwarz divergence.

Impact of the Hyperparameter λ:

The coefficient λ controls the relative strength of the prior-matching constraint when using the Cauchy-Schwarz metric instead of the standard KL divergence. This modification has several important implications:

- Since we proved that $D_{CS}(q||p) \leq D_{KL}(q||p)$, the gradient signals provided by the Cauchy-Schwarz divergence are generally more bounded and smoother than those from the KL divergence. This can lead to more stable training dynamics.

- Increasing λ emphasizes the Cauchy-Schwarz metric as the primary regularizer. The Cauchy-Schwarz divergence is derived from the Rényi divergence of order 2, which penalizes the distributional overlap differently compared to KL divergence (which is based on logarithmic ratios).

- From a qualitative perspective, the KL divergence exhibits "mode-covering" behavior, forcing the approximate posterior q to place probability mass wherever the prior p has support, and heavily penalizing q for having mass where p is zero. In contrast, the Cauchy-Schwarz divergence D_{CS} is more tolerant of such discrepancies.

- A higher value of λ combined with the use of D_{CS} allows the learned latent representation to deviate more freely from the prior without incurring severe penalties. This greater flexibility can potentially prevent the "posterior collapse" problem—where the model ignores the latent variables and the approximate posterior becomes identical to the prior—and enables the learning of more expressive and structured latent representations. The penalty for deviating from the prior is less aggressive compared to the standard KL-based regularization term.

In summary, the modified objective can be expressed as:

$$\mathcal{L}_{CS} = \log p_\theta(x) - D_{KL}(q_\phi(z|x)||p_\theta(z|x)) + (1-\lambda)D_{KL}(q||p) + \lambda(D_{KL}(q||p) - D_{CS}(q||p))$$

This formulation trades off reconstruction accuracy, posterior approximation quality, and a softened prior-matching constraint controlled by λ and the Cauchy-Schwarz divergence.

Problem 3: Posterior Collapse in Variational Autoencoders

1. Description of Posterior Collapse

Posterior collapse (also referred to as KL collapse or vanishing KL) is the regime where the latent variable z of a VAE effectively stops carrying information about the observation x. In such a situation the decoder $p_\theta(x \mid z)$ is so expressive that it can reproduce the data distribution without relying on z. As a result, the encoder has little incentive to encode anything and simply outputs the prior.

A collapsed solution can be summarized as

$$\forall x \sim p_{\text{data}}(x), \qquad q_\phi(z \mid x) \approx p(z)$$

A global view uses the expected KL divergence between posterior and prior:

$$\mathbb{E}_{x \sim p_{\text{data}}(x)} \Big[D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big) \Big] \approx 0$$

Equivalently, collapse corresponds to near–zero mutual information between x and z under the encoder joint distribution:

$$I_q(x; z) = \mathbb{E}_{q(x,z)} \left[\log \frac{q(x, z)}{p_{\text{data}}(x)\, q_\phi(z)} \right] \approx 0,$$

where

$$q(x, z) = p_{\text{data}}(x)\, q_\phi(z \mid x), \qquad q_\phi(z) = \int p_{\text{data}}(x)\, q_\phi(z \mid x)\, dx.$$

2. Optimal Encoder with an Overly Expressive Decoder

Assume the decoder can model the data without using z; i.e. there exists a distribution $\tilde{p}(x)$ such that

$$\forall z, \qquad p_\theta(x \mid z) = \tilde{p}(x) \approx p_{\text{data}}(x).$$

For a single data point x, the ELBO is

$$\text{ELBO}(x; \theta, \phi) = \mathbb{E}_{z \sim q_\phi(z|x)} \big[\log p_\theta(x \mid z)\big] - D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big).$$

Substitute the decoder independence:

$$\mathbb{E}_{z \sim q_\phi(z|x)} \big[\log p_\theta(x \mid z)\big] = \mathbb{E}_{z \sim q_\phi(z|x)} \big[\log \tilde{p}(x)\big] = \log \tilde{p}(x),$$

since $\log \tilde{p}(x)$ does not depend on z. Thus,

$$\text{ELBO}(x; \theta, \phi) = \log \tilde{p}(x) - D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big).$$

When optimizing the encoder parameters ϕ, the term $\log \tilde{p}(x)$ is constant, so

$$\max_\phi \text{ELBO}(x; \theta, \phi) \iff \min_\phi D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big).$$

By Gibbs' inequality,

$$D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big) \geq 0, \quad D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big) = 0 \iff q_\phi(z \mid x) = p(z).$$

Hence, in this regime the best encoder ignores x entirely and outputs the prior, which is precisely posterior collapse.

3. Decomposition of the Expected KL Term

Consider the ELBO averaged over the data distribution:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{x \sim p_{\text{data}}(x)} \Big[\mathbb{E}_{z \sim q_\phi(z|x)} \big[\log p_\theta(x \mid z) \big] - D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z) \big) \Big].$$

Define the joint and the aggregated posterior as

$$q(x, z) = p_{\text{data}}(x)\, q_\phi(z \mid x), \qquad q_\phi(z) = \int p_{\text{data}}(x)\, q_\phi(z \mid x)\, dx.$$

The expected KL regularizer can then be written as

$$\begin{aligned}
\mathbb{E}_{p_{\text{data}}(x)} \big[D_{\text{KL}}(q_\phi(z \mid x) \| p(z)) \big] &= \int p_{\text{data}}(x) \int q_\phi(z \mid x) \log \frac{q_\phi(z \mid x)}{p(z)}\, dz\, dx \\
&= \iint q(x, z) \log \frac{q_\phi(z \mid x)}{p(z)}\, dz\, dx.
\end{aligned}$$

Introduce $q_\phi(z)$ inside the logarithm:

$$\begin{aligned}
\iint q(x, z) \log \frac{q_\phi(z \mid x)}{p(z)}\, dz\, dx &= \iint q(x, z) \log \left(\frac{q_\phi(z \mid x)}{q_\phi(z)} \cdot \frac{q_\phi(z)}{p(z)} \right) dz\, dx \\
&= \iint q(x, z) \log \frac{q_\phi(z \mid x)}{q_\phi(z)}\, dz\, dx + \iint q(x, z) \log \frac{q_\phi(z)}{p(z)}\, dz\, dx.
\end{aligned}$$

The first integral is the mutual information $I_q(x; z)$:

$$\iint q(x, z) \log \frac{q_\phi(z \mid x)}{q_\phi(z)}\, dz\, dx = \iint q(x, z) \log \frac{q(x, z)}{p_{\text{data}}(x)\, q_\phi(z)}\, dz\, dx = I_q(x; z).$$

For the second integral, integrate out x:

$$\begin{aligned}
\iint q(x, z) \log \frac{q_\phi(z)}{p(z)}\, dz\, dx &= \int \left[\log \frac{q_\phi(z)}{p(z)} \int q(x, z)\, dx \right] dz = \int q_\phi(z) \log \frac{q_\phi(z)}{p(z)} dz \\
&= D_{\text{KL}}\big(q_\phi(z) \,\|\, p(z) \big).
\end{aligned}$$

Thus the expected KL splits into a mutual-information term and a KL between aggregated posterior and prior:

$$\mathbb{E}_{p_{\text{data}}(x)} \big[D_{\text{KL}}(q_\phi(z \mid x) \| p(z)) \big] = I_q(x; z) + D_{\text{KL}}\big(q_\phi(z) \,\|\, p(z) \big).$$

Substituting into the dataset ELBO yields

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{p_{\text{data}}(x)} \mathbb{E}_{q_\phi(z|x)} \big[\log p_\theta(x \mid z) \big] - D_{\text{KL}}\big(q_\phi(z) \,\|\, p(z) \big) - I_q(x; z).$$

In a fully collapsed model we have approximately $I_q(x; z) = 0$, so x and z are nearly independent under the learned joint.

4. Common Causes of Collapse

Several modelling and optimization choices tend to push the VAE towards the collapsed regime:

- **Very expressive autoregressive decoders.** Strong autoregressive architectures (e.g. powerful CNNs, Transformers) may reconstruct x accurately from its own history, making z redundant.
- **Aggressive KL regularization.** A large weight on the KL term at the beginning of training can force $q_\phi(z \mid x)$ to match the prior before useful structure is encoded in z.
- **Lagging inference networks.** When the decoder learns much faster than the encoder, it can quickly fit the data while the encoder produces noisy codes; the easiest way for the decoder to reduce loss is then to ignore z.

5. Strategies for Mitigating Posterior Collapse

Practical techniques to maintain informative latent variables include:

- **KL annealing.** Use a time–varying coefficient β on the KL term:

  $$\mathcal{L}_\beta = \mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] - \beta\, D_{\mathrm{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big), \qquad \beta : 0 \to 1.$$

  Early on, β is small so the model mainly learns reconstruction; later, the KL term is gradually enforced.

- **Free bits / KL thresholding.** Penalize only the portion of the KL exceeding a target value λ:

  $$\mathcal{L}_{\mathrm{FB}} = \mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] - \max\big(\lambda,\ D_{\mathrm{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big)\big)\,.$$

  This keeps at least λ nats/bits of capacity available for information in z.

- **Weakening the decoder.** Use a less powerful or more regularized decoder so that good reconstructions require informative latent codes; this can be done via smaller networks, word dropout, or limited receptive fields.

$$\boxed{\mathcal{L}(\theta, \phi) = \mathbb{E}_{p_{\mathrm{data}}(x)}\mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] - D_{\mathrm{KL}}\big(q_\phi(z) \,\|\, p(z)\big) - I_q(x; z)}$$

with posterior collapse when $I_q(x; z) \approx 0$.

Problem 4: Probabilistic Graph Forecasting with Autoregressive Decoders

1. Joint Factorization and Sequence ELBO

We observe a sequence of graphs $(X_{1:T}, A_{1:T}) = \{(X_1, A_1), \ldots, (X_T, A_T)\}$ and introduce latent states $Z_{0:T} = \{Z_0, \ldots, Z_T\}$. The generative model specifies a prior over the initial state and, for each time step, a transition and an emission (decoder):

$$p_\theta(X_{1:T}, A_{1:T}, Z_{0:T}) = p(Z_0) \prod_{t=0}^{T-1} \Big[p_\theta(Z_{t+1} \mid Z_t)\, p_\theta(X_{t+1}, A_{t+1} \mid Z_t)\Big].$$

Our training objective is the log marginal likelihood of the observed graphs:

$$\log p_\theta(X_{1:T}, A_{1:T}) = \log \int p_\theta(X_{1:T}, A_{1:T}, Z_{0:T})\, dZ_{0:T}.$$

Because this integral is intractable, we introduce a structured approximate posterior over the future latents $Z_{1:T}$:

$$q_\phi(Z_{1:T} \mid X_{1:T}, A_{1:T}) = \prod_{t=0}^{T-1} q_\phi\big(Z_{t+1} \mid Z_t, X_{1:t+1}, A_{1:t+1}\big),$$

which can be interpreted as a recursive inference model (e.g. a GNN running forward in time).

Insert this distribution inside the marginal and apply Jensen's inequality:

$$\begin{aligned}\log p_\theta(X_{1:T}, A_{1:T}) &= \log \int q_\phi(Z_{1:T} \mid X_{1:T}, A_{1:T})\, \frac{p_\theta(X_{1:T}, A_{1:T}, Z_{0:T})}{q_\phi(Z_{1:T} \mid X_{1:T}, A_{1:T})}\, dZ_{0:T} \\ &\geq \mathbb{E}_{q_\phi(Z_{1:T}\mid X_{1:T}, A_{1:T})}\left[\log \frac{p(Z_0)\prod_{t=0}^{T-1} p_\theta(Z_{t+1} \mid Z_t)\, p_\theta(X_{t+1}, A_{t+1} \mid Z_t)}{\prod_{t=0}^{T-1} q_\phi(Z_{t+1} \mid Z_t, X_{1:t+1}, A_{1:t+1})}\right].\end{aligned}$$

Expanding the logarithm of the product into a sum and using linearity of expectation yields

$$\begin{aligned}\log p_\theta(X_{1:T}, A_{1:T}) \;\geq\; \mathbb{E}_{q_\phi(Z_{1:T}\mid X_{1:T}, A_{1:T})}\Bigg[&\log p(Z_0) + \sum_{t=0}^{T-1} \log p_\theta(X_{t+1}, A_{t+1} \mid Z_t) \\ &+ \sum_{t=0}^{T-1} \log p_\theta(Z_{t+1} \mid Z_t) - \sum_{t=0}^{T-1} \log q_\phi\big(Z_{t+1} \mid Z_t, X_{1:t+1}, A_{1:t+1}\big)\Bigg].\end{aligned}$$

Assuming the initial state Z_0 is fixed or drawn from a prior that is not optimized, the term $\log p(Z_0)$ can be dropped as a constant. Collecting terms for each time step t, and abbreviating $q_\phi(Z_t \mid X_{1:t}, A_{1:t})$ as the corresponding filtering distribution, we obtain the sequence ELBO

$$\begin{aligned}\mathcal{L}(\theta, \phi) = \sum_{t=0}^{T-1} \mathbb{E}_{q_\phi(Z_t\mid X_{1:t}, A_{1:t})}\Big[&\log p_\theta(X_{t+1}, A_{t+1} \mid Z_t) \\ &- D_{\mathrm{KL}}\Big(q_\phi(Z_{t+1} \mid Z_t, X_{1:t+1}, A_{1:t+1}) \,\|\, p_\theta(Z_{t+1} \mid Z_t)\Big)\Big].\end{aligned}$$

The first term at each t is a reconstruction term for the next graph, while the second is a temporal KL divergence regularizing the transition posterior towards the latent prior. For $t = 0$, the expectation is trivial because Z_0 is fixed, so the reconstruction part reduces to $\log p_\theta(X_1, A_1 \mid Z_0)$ and the KL compares $q_\phi(Z_1 \mid Z_0, X_1, A_1)$ against the prior transition $p_\theta(Z_1 \mid Z_0)$.

2. Node and Edge Reconstruction Likelihoods

Given a latent state Z_t, the decoder specifies distributions for both node features and edges at the next time step.

(a) Node feature term (Gaussian). Let x_v^{t+1} be the feature vector of node v at time $t+1$, and let the decoder output a mean $\mu_v^\theta(Z_t)$ and covariance $\Sigma_v^\theta(Z_t)$. Assuming conditional independence across nodes, the Gaussian log-likelihood is

$$\log p_\theta(X_{t+1} \mid Z_t) = \sum_{v=1}^{n} \Bigg[-\frac{1}{2}\big(x_v^{t+1} - \mu_v^\theta(Z_t)\big)^\top \big(\Sigma_v^\theta(Z_t)\big)^{-1} \big(x_v^{t+1} - \mu_v^\theta(Z_t)\big) \\ -\frac{1}{2}\log\big|2\pi\, \Sigma_v^\theta(Z_t)\big| \Bigg].$$

(b) Adjacency term (Bernoulli). For each unordered pair of nodes (u, v) with $u < v$, the edge indicator A_{uv}^{t+1} is modeled as a Bernoulli with parameter $\pi_{uv}^\theta(Z_t)$:

$$\log p_\theta(A_{t+1} \mid Z_t) = \sum_{u<v} \Big[A_{uv}^{t+1} \, \log \pi_{uv}^\theta(Z_t) + \big(1 - A_{uv}^{t+1}\big) \log\big(1 - \pi_{uv}^\theta(Z_t)\big) \Big].$$

The total reconstruction contribution at time t, entering the ELBO above, is the sum $\log p_\theta(X_{t+1} \mid Z_t) + \log p_\theta(A_{t+1} \mid Z_t)$.

3. Deterministic Decoder: Consequences

If the decoder is made deterministic, for example by driving the Gaussian covariance $\Sigma_v^\theta(Z_t)$ towards zero for all nodes, the likelihood becomes extremely peaked.

Effect on the ELBO. In the limit $\Sigma \to 0$ the Gaussian density approaches a delta function: perfect reconstructions yield arbitrarily large log-likelihood, while even tiny errors give very negative values. Ignoring constants, the reconstruction term behaves like a squared error loss, and the probabilistic interpretation of the ELBO is lost.

Loss of multimodality. A deterministic mapping $Z_t \mapsto (X_{t+1}, A_{t+1})$ is one-to-one, so the model cannot represent genuinely multimodal futures. If the next graph could evolve in several distinct plausible ways given Z_t, a deterministic decoder tends to predict an "average" graph, which may correspond to no realistic outcome.

Stability issues. Eliminating variance removes an important mechanism for handling uncertainty and noise. The model may overfit to particular training trajectories, leading to mode collapse or unstable gradients as it tries to fit noisy details exactly.

4. Generation Procedure and Long-Term Behaviour

Suppose we have observed a history $(X_{1:T}, A_{1:T})$ and inferred the final latent state Z_T. To forecast future graphs for a horizon of H steps, we simulate the latent Markov chain and decoder autoregressively.

Rollout algorithm. Initialize the current latent to the last inferred state, $\hat{Z}_T = Z_T$. For $k = 0, 1, \ldots, H-1$:

- **Generate graph:** sample the next graph $(\hat{X}_{T+k+1}, \hat{A}_{T+k+1}) \sim p_\theta(X, A \mid \hat{Z}_{T+k})$.
- **Update latent:** sample the next latent state $\hat{Z}_{T+k+1} \sim p_\theta(Z \mid \hat{Z}_{T+k})$.

Convergence conditions. Consider the latent transition kernel $p_\theta(Z_{t+1} \mid Z_t)$. If this Markov chain is *ergodic* (irreducible and aperiodic with non-zero transition probabilities), the distribution of Z_t converges to a unique stationary distribution $\pi(Z)$ as $t \to \infty$. This distribution satisfies the fixed-point integral equation

$$\pi(Z') = \int p_\theta(Z' \mid Z)\, \pi(Z)\, dZ.$$

Consequently, after sufficient steps, generated graphs are distributed according to the marginal obtained by passing $\pi(Z)$ through the decoder:

$$p_\infty(X, A) = \int p_\theta(X, A \mid Z)\, \pi(Z)\, dZ.$$

5. Amortization Gap and Distribution Drift

Let $p_\theta(Z_t \mid X_{1:t})$ denote the true but intractable posterior over Z_t given the observed prefix, and $q_\phi(Z_t \mid X_{1:t})$ the approximate posterior produced by the inference network. The *amortization gap* at time t is

$$\Delta_{\text{gap}} = D_{\text{KL}}\big(q_\phi(Z_t \mid X_{1:t}) \,\|\, p_\theta(Z_t \mid X_{1:t})\big) > 0.$$

This quantity is generally strictly positive, meaning that even with perfect training of the inference network, the posterior samples used during learning do not perfectly match the true posterior.

This leads to a covariate-shift phenomenon between training and generation:

- **Training phase (teacher forcing).** The transition prior $p_\theta(Z_{t+1} \mid Z_t)$ is trained on latent inputs Z_t sampled from the encoder, $Z_t \sim q_\phi(\cdot \mid X_{1:t})$.
- **Generation phase (autoregressive).** At test time, the model conditions on its own previous latent predictions, so Z_t is drawn from the learned transition kernel, approximately $Z_t \sim p_\theta(\cdot \mid \hat{Z}_{t-1})$.

Because the transition model is optimized under one input distribution but used under another, small discrepancies can accumulate over time.

6. Error Propagation and Scheduled Sampling Objective

Suppose that at some step t the predicted latent $\hat{Z}_t$ is slightly erroneous relative to the "ideal" latent Z_t, with $\|\hat{Z}_t - Z_t\| \approx \varepsilon$. If the transition dynamics are Lipschitz with constant $L > 1$ (typical in unstable or chaotic dynamics), the prediction error at the next step satisfies

$$\|\hat{Z}_{t+1} - Z_{t+1}\| \;\leq\; L\,\|\hat{Z}_t - Z_t\| + \text{noise} \;\approx\; L\,\varepsilon.$$

After k rollout steps the discrepancy can grow roughly as $L^k \varepsilon$, causing the latent trajectory to drift away from the manifold on which the decoder was trained, resulting in poor graph forecasts.

Scheduled sampling. To reduce this train–test mismatch we mix teacher-forced and autoregressive inputs during training. At training iteration i we draw a Bernoulli variable $c_t \sim \text{Bernoulli}(\epsilon_i)$ and define the latent input $\tilde{Z}_t$ to the transition model as a mixture of encoder and model samples:

$$\tilde{Z}_t = \begin{cases} Z_t \sim q_\phi(Z_t \mid X_{1:t}), & \text{with probability } \epsilon_i \quad \text{(teacher forcing)}, \\ \hat{Z}_t \sim p_\theta(Z_t \mid \hat{Z}_{t-1}), & \text{with probability } 1 - \epsilon_i \quad \text{(autoregressive)}. \end{cases}$$

$$\text{Training objective:} \qquad \max_\theta \; \mathbb{E}_{\tilde{Z}_t}\big[\log p_\theta(Z_{t+1} \mid \tilde{Z}_t)\big].$$

Here ϵ_i is annealed from 1 (pure teacher forcing) to 0 (fully autoregressive) over the course of training. By repeatedly training on its own noisy latent predictions $\tilde{Z}_t$, the transition model learns to correct small deviations and steer the trajectory back towards the manifold of valid latent states, thereby mitigating long-term drift.